

Column Formwork Fabrication Guide

v5.0 Ultra-Optimised
(Asymmetric Corners)

COLUMN DIMENSIONS: 450 × 300 MM (12 × 18 IN) |
HEIGHT: 8 FT (2438 MM)

Fabricator Summary: This guide provides a clean, concise reference for the site welding team. It isolates the **Cut List**, **Material Weights**, and the **Step-by-Step Build Schedule** with English-labeled engineering diagrams.

1. Total Material Requirements				
Material	Size / Specification	Quantity	Total Weight	Notes
MS Steel Angles	ISA 35 × 35 × 5 mm	8 bars × 20 ft	112.7 kg	Use remnants/offcuts for kickers & bracing
MS Flat Bar	75 × 5 mm flat bar	2 bars × 8 ft	14.4 kg	Centre-line stiffener for the 300 mm faces
Fasteners (High-Grade)	M16 Grade 8.8 bolts/nuts	16 sets	~2.5 kg	Use on horizontal Yokes 1, 2, 3, 4, 5 (base pressure zone)
Fasteners (Standard)	M12 Grade 4.6 bolts/nuts	100+ sets	~3.0 kg	For plywood sandwich clamping and Yokes 6, 7, 8, 9
Shuttering Plywood	12 mm BWP Grade 1 (IS 4990)	4 panels	—	2 panels: 508 mm (20") wide, 2 panels: 300 mm wide
Total Steel Weight per Set			~139.1 kg	Weight reduced by 22 kg (vs v4.0) and 60 kg (vs v3.0) ✓

2. Cut List & Sourcing

A. Vertical Frame Members (ISA 35×35×5)

Component	Length	Qty (Singles)	Pairs (Welded)	Description
Double corners (TL & BR)	2 438 mm (8 ft)	4	2 pairs	Welded back-to-back
Single corners (TR & BL)	2 438 mm (8 ft)	2	—	Keep as single angles (unwelded)
Central stringers (450mm face)	2 438 mm (8 ft)	2	—	Keep as single angles (unreinforced)

B. Horizontal Yokes / Clamps (ISA 35×35×5)

Yoke Component	Length	For Single Yokes	For Double Yokes	Total Singles
Long sides	550 mm	10 pcs	16 pcs	26 pcs
Short sides	400 mm	10 pcs	16 pcs	26 pcs

3. Yoke Spacing Schedule (Heights Measured from Ground)

Yoke 1 anchors at ground level (0 mm) and Yoke 9 clamps the top crown (2438 mm). Spacing graduates wider toward the top as pressure drops.

Yoke #	Height from Base	Interval (Gap)	Yoke Configuration	Required Corner Bolt
Yoke 1 (Base)	0 mm	0 mm	Single Angle (ISA 35×5)	M16 Grade 8.8
Yoke 2	200 mm	200 mm	Double Angle (ISA 35×5)	M16 Grade 8.8
Yoke 3	420 mm	220 mm	Double Angle (ISA 35×5)	M16 Grade 8.8
Yoke 4	660 mm	240 mm	Double Angle (ISA 35×5)	M16 Grade 8.8
Yoke 5	920 mm	260 mm	Double Angle (ISA 35×5)	M16 Grade 8.8
Yoke 6	1200 mm	280 mm	Single Angle (ISA 35×5)	M12 Grade 4.6
Yoke 7	1500 mm	300 mm	Single Angle (ISA 35×5)	M12 Grade 4.6
Yoke 8	1870 mm	370 mm	Single Angle (ISA 35×5)	M12 Grade 4.6
Yoke 9 (Top)	2438 mm	568 mm	Single Angle (ISA 35×5)	M12 Grade 4.6

4. Step-by-Step Build Schedule

1 Marking and Cutting

Cut all angles and plates per the cut list. Mark and pre-drill all M12 and M16 bolt holes before welding to ensure accuracy.

2 Welding Vertical Corner Members

Weld 2 pairs of corner verticals back-to-back (TL and BR corners only) using a continuous/intermittent fillet weld (≥ 5 mm). TR and BL corners, and the central stringers, must remain as single angles.

3 Weld Vertical Stiffeners to Yokes

Fit the two 75×5 mm vertical flat bars down the centerline of the 300 mm faces and weld them securely at each yoke intersection. This creates a vertical beam support structure.

4 Assemble the 450 mm Face Panels

Sandwich-bolt the 20-inch (508 mm) plywood sheathing panels to the central stringers and double corner angles. Spacing of M12 bolts should be ≤ 250 mm.

5 Erect Panels and Clamp Yokes (Base to Top)

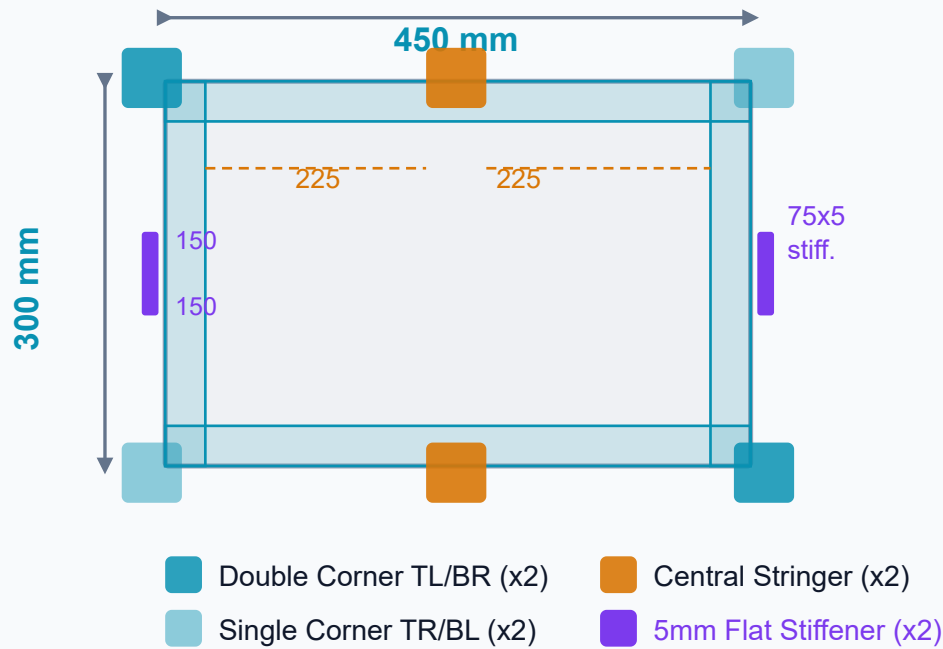
Erect the panels, verify plumb on both axes, and secure with diagonal kicker braces. Fit yokes starting from the ground (0 mm) up to 2438 mm. Use **M16 Grade 8.8 bolts** on Yokes 1 to 5.

6 Pre-Pour Checks and Concreting

Verify all fasteners are tight and seams are sealed. Pour M25 concrete in lifts of no more than 500 mm, compacting with a vibrator between each lift.

5. Expressive Assembly Diagrams

CROSS-SECTION (TOP-DOWN VIEW)



ELEVATION (VERTICAL ASSEMBLY)

